

CMOS VLSI Design Project

Design and Analysis of a 2-Input CMOS XOR Gate

Using Cadence Virtuoso

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1 Introduction

The CMOS XOR gate is an important building block of digital integrated circuits. It performs the exclusive-OR logical operation, where the output becomes HIGH when the two input signals are different.

In this project, a two-input CMOS XOR gate was designed and analyzed using the Cadence Virtuoso design environment. The complete custom CMOS design flow included schematic design, symbol generation, physical layout design, Design Rule Check (DRC), Layout Versus Schematic (LVS) verification, RC parasitic extraction (RCX), and post-layout simulation.

The post-layout simulation was used to evaluate important performance parameters such as average power consumption and propagation delay.

The Boolean expression of a two-input XOR gate is

$$Y = A \oplus B \quad (1)$$

which can also be expressed as

$$Y = \overline{A}B + A\overline{B}. \quad (2)$$

Here, A and B represent the input signals and Y represents the output.

2 Objectives

The main objectives of this project are:

- To design a two-input CMOS XOR gate using Cadence Virtuoso.
- To create and verify the transistor-level schematic of the XOR gate.
- To generate a reusable symbol for the designed XOR gate.
- To design the physical layout of the CMOS XOR gate.
- To verify the physical layout using Design Rule Check (DRC).
- To verify the electrical equivalence between the schematic and layout using Layout Versus Schematic (LVS).
- To perform parasitic resistance and capacitance extraction using RCX.
- To perform post-layout transient simulation using the extracted circuit.
- To calculate the average power consumption of the XOR gate.
- To calculate the propagation delay of the XOR gate.

3 Theory of CMOS XOR Gate

A two-input XOR gate performs the exclusive-OR Boolean operation. The output is logic HIGH when the two input signals are different and logic LOW when the two input signals are the same.

The Boolean expression of the XOR gate is

$$Y = A \oplus B. \quad (3)$$

The XOR operation can also be written in sum-of-products form as

$$Y = \overline{A}B + A\overline{B}. \quad (4)$$

The truth table of the two-input CMOS XOR gate is shown in Table 1.

Table 1: Truth Table of a Two-Input CMOS XOR Gate

A	B	Y
0	0	0
0	1	1
1	0	1
1	1	0

When both inputs are LOW, the XOR output is LOW because the two input signals are equal.

When $A = 0$ and $B = 1$, the inputs are different and the output becomes HIGH.

Similarly, when $A = 1$ and $B = 0$, the inputs are different and the output becomes HIGH.

When both inputs are HIGH, the inputs are again equal and the output becomes LOW.

Therefore, the required XOR function is

$$Y = \overline{A}B + A\overline{B}. \quad (5)$$

4 Transistor Sizing

The CMOS XOR gate was designed using the required PMOS and NMOS transistors according to the transistor-level implementation in Cadence Virtuoso.

The transistor dimensions used in the designed XOR circuit should be reported according to the actual Cadence schematic.

Table 2: Transistor Dimensions Used in the CMOS XOR Gate

Transistor	Number	Width	Length
PMOS	2	480 nm	100 nm
NMOS	2	240 nm	100 nm

The transistor sizing was selected according to the required CMOS implementation and the design parameters of the selected technology.

5 Circuit Design

5.1 Schematic Design

The two-input CMOS XOR gate schematic was created using the Cadence Virtuoso Schematic Editor.

The transistor-level circuit was constructed to realize the Boolean function

$$Y = A \oplus B. \quad (6)$$

The two input terminals are represented by A and B , while the final output terminal is represented by Y .

The schematic of the designed XOR gate is shown in Figure 1.

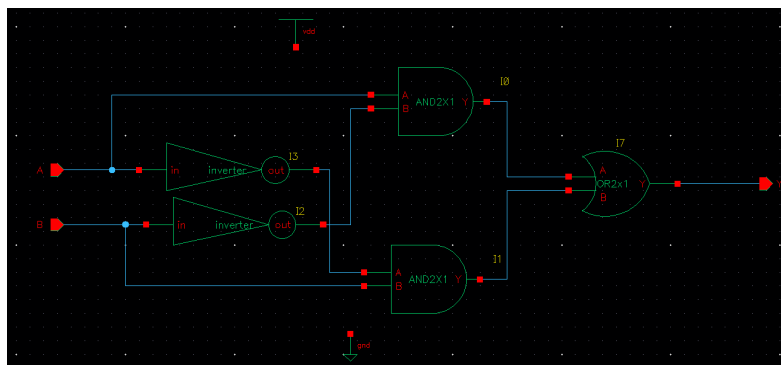


Figure 1: Schematic of the two-input CMOS XOR gate in Cadence Virtuoso.

The logical operation of the circuit can be represented as

$$A, B \rightarrow A \oplus B. \quad (7)$$

5.2 XOR Gate Symbol

After completing the schematic, a symbol was generated for the two-input CMOS XOR gate.

The symbol contains two input terminals labeled A and B and one output terminal labeled Y .

The generated XOR gate symbol is shown in Figure 2.

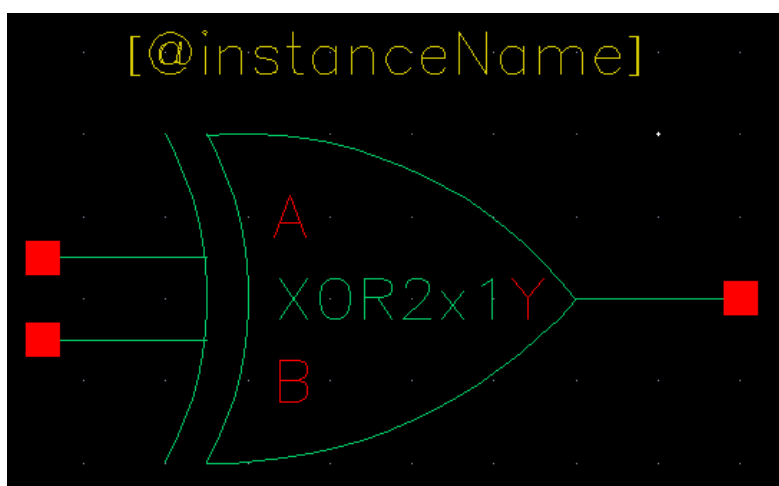


Figure 2: Generated symbol of the two-input CMOS XOR gate.

The symbol allows the designed XOR gate to be used conveniently as a reusable hierarchical block in larger digital circuits.

6 Physical Layout Design

After completing the schematic and symbol generation, the physical layout of the CMOS XOR gate was developed using Cadence Virtuoso Layout Editor.

The layout represents the physical implementation of the transistor devices and their electrical interconnections.

The physical layout contains the required layers for transistor formation and electrical interconnection, including:

- active or diffusion regions,
- polysilicon,
- N-well,
- metal layers,
- contacts,
- vias,

- power connection, and
- ground connection.

The input signals are routed to the appropriate transistor gates and the transistor networks are interconnected to realize the XOR operation.

The completed physical layout is shown in Figure 3.

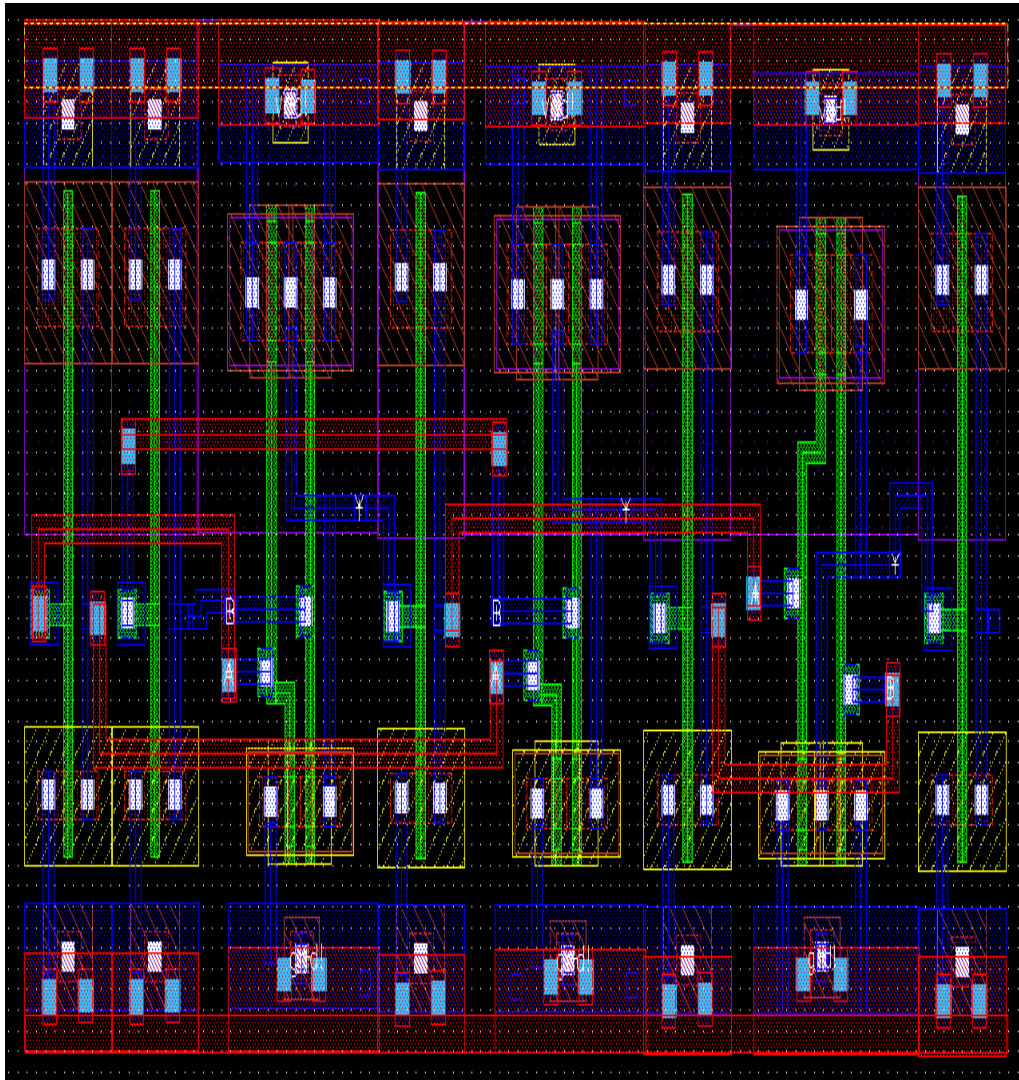


Figure 3: Physical layout of the two-input CMOS XOR gate.

The layout was designed while maintaining the required minimum width, spacing, enclosure, and connectivity rules of the selected CMOS technology.

7 Design Rule Check (DRC)

Design Rule Check (DRC) is performed to verify whether the physical layout satisfies the fabrication rules of the selected CMOS technology.

DRC checks various geometrical constraints such as:

- minimum metal width,
- minimum metal spacing,
- minimum polysilicon width,
- polysilicon spacing,
- diffusion spacing,
- contact dimensions,
- contact enclosure,
- via enclosure, and
- well spacing.

The completed XOR gate layout was checked using the Cadence DRC verification flow.

DRC Result: PASSED

The physical layout satisfies the required design rules of the selected CMOS technology.

Thus, the physical layout is valid with respect to the required fabrication design rules.

8 Layout Versus Schematic (LVS)

After successful DRC verification, Layout Versus Schematic (LVS) verification was performed.

LVS compares the electrical netlist extracted from the physical layout with the netlist generated from the original schematic.

It verifies important circuit properties such as:

- number of MOS transistors,
- PMOS and NMOS device types,
- gate connectivity,
- source and drain connectivity,
- input connections,
- output connection,
- power connection, and
- ground connection.

The final XOR gate layout was checked against the original schematic.

LVS Result: PASSED

The physical layout is electrically equivalent to the original schematic.

The successful LVS result confirms that the intended electrical connectivity was maintained during the physical layout design.

9 RCX and Parasitic Extraction

After DRC and LVS verification, RC parasitic extraction was performed using the RCX extraction flow.

Physical interconnections in an integrated circuit introduce additional parasitic resistance and capacitance.

These parasitic elements can affect important performance parameters such as propagation delay and power consumption.

RCX extracts these parasitic resistance and capacitance components from the physical layout.

The extracted view of the designed XOR gate is shown in [Figure 4](#).

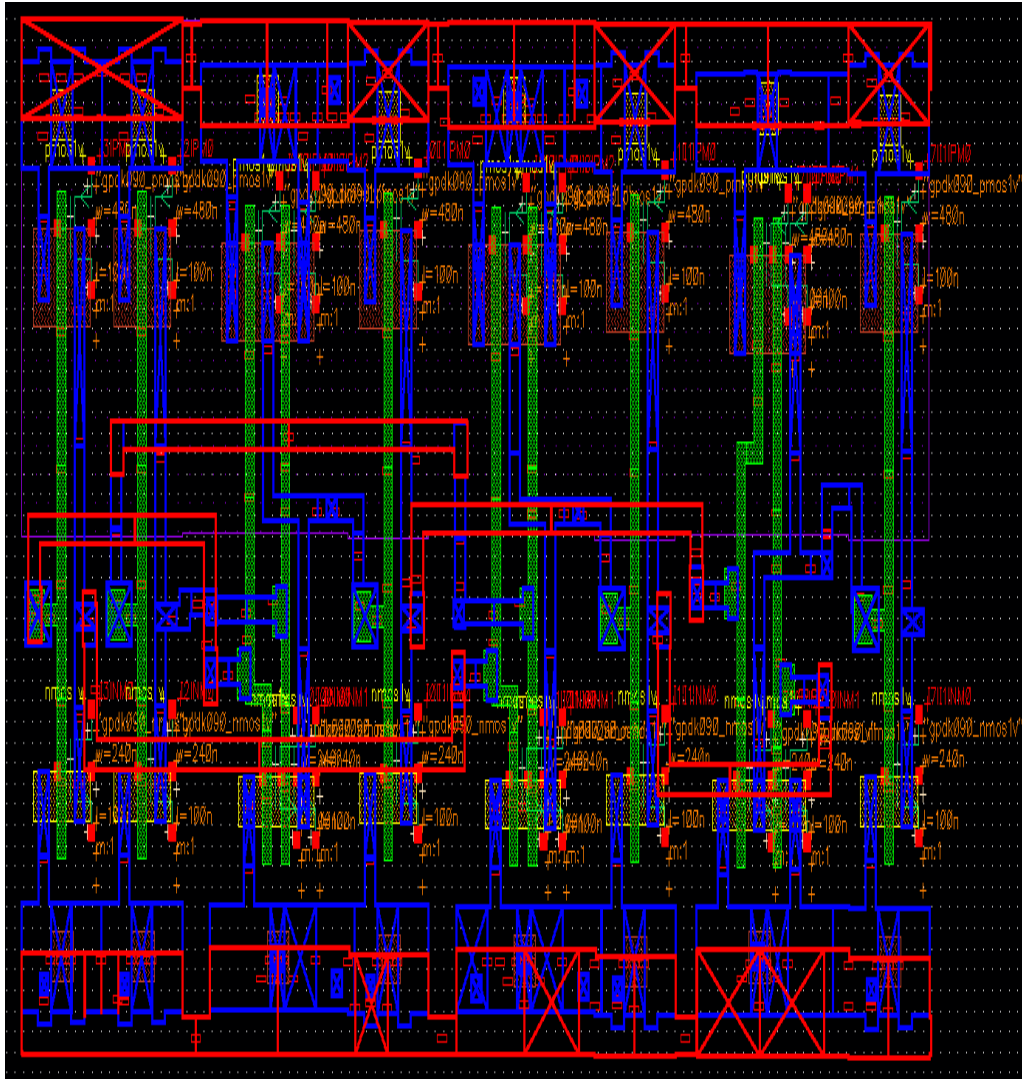


Figure 4: RCX/extracted view of the two-input CMOS XOR gate.

RCX: Successfully Completed

The extracted XOR gate circuit was used for post-layout simulation.

10 Post-Layout Simulation

After parasitic extraction, post-layout transient simulation was performed using the RCX-extracted view of the XOR gate.

Unlike ideal schematic-level simulation, post-layout simulation includes the effects of parasitic resistance and capacitance generated by the physical layout.

Therefore, post-layout simulation provides a more realistic estimation of the circuit performance.

The main performance parameters calculated in this project were:

1. Average power consumption

2. Propagation delay

11 Average Power Calculation

The average power consumption of the XOR gate was determined from the post-layout transient simulation results using the Cadence Virtuoso Calculator.

The average power over a time interval T can be expressed as

$$P_{\text{avg}} = \frac{1}{T} \int_0^T p(t) dt \quad (8)$$

where $p(t)$ represents the instantaneous power consumption.

The average value of the simulated power waveform was calculated using the Cadence Virtuoso Calculator.

The power calculation is shown in Figure 5.



Figure 5: Average power calculation of the extracted CMOS XOR gate using the Cadence Virtuoso Calculator.

From the Cadence Calculator, the obtained average power is

$$P_{\text{avg}} = 13.2 \times 10^{-6} \text{ W}. \quad (9)$$

Since

$$1 \mu\text{W} = 10^{-6} \text{ W}, \quad (10)$$

the average power consumption can be written as

$$P_{\text{avg}} = 13.2 \mu\text{W} \quad (11)$$

Therefore, the post-layout average power consumption of the designed two-input CMOS XOR gate is approximately

$$13.2 \mu\text{W}. \quad (12)$$

12 Propagation Delay Calculation

Propagation delay represents the amount of time required for an input transition to produce the corresponding transition at the output of a digital circuit.

For the XOR gate, a selected input transition that causes the output to change was used to determine the propagation delay.

The propagation delay can generally be expressed as

$$t_p = t_{\text{out}} - t_{\text{in}}. \quad (13)$$

In this simulation, a switching reference level of approximately 0.6 V was used.

The Cadence Virtuoso Calculator was used to determine the delay between the selected input transition and the corresponding output transition.

The delay calculation is shown in Figure 6.



Figure 6: Propagation-delay calculation of the extracted CMOS XOR gate using the Cadence Virtuoso Calculator.

The measured value displayed by the Cadence Calculator is

$$t_p = -9.757 \times 10^{-9} \text{ s}. \quad (14)$$

Since

$$1 \text{ ns} = 10^{-9} \text{ s}, \quad (15)$$

the displayed propagation-delay result becomes

$$t_p = -9.757 \text{ ns} \quad (16)$$

The Calculator therefore reports a negative value for the selected input-output transition.

A negative propagation delay is not the usual physical interpretation of a digital gate delay and may indicate that the selected threshold crossings or transition directions in the Calculator expression are reversed.

Therefore, the displayed value is reported here exactly as obtained from the Cadence Calculator.

13 Verification and Performance Summary

The overall design, verification, and performance results of the two-input CMOS XOR gate are summarized in Table 3.

Table 3: Summary of the CMOS XOR Gate Design Results

Parameter	Result
Logic Function	$Y = A \oplus B$
Boolean Expression	$Y = \overline{A}B + A\overline{B}$
Transistor Dimensions	According to the designed XOR schematic
Schematic Design	Completed
Symbol Generation	Completed
Physical Layout	Completed
DRC	Passed
LVS	Passed
RCX / Parasitic Extraction	Completed successfully
Post-layout Simulation	Completed
Average Power	13.2 μ W
Propagation Delay	−9.757 ns

14 Discussion

The complete custom CMOS design flow for a two-input XOR gate was carried out using Cadence Virtuoso.

The XOR gate performs the exclusive-OR operation, producing a logic HIGH output when the two input signals are different and a logic LOW output when they are equal.

The required Boolean function is

$$Y = A \oplus B \quad (17)$$

or equivalently,

$$Y = \overline{A}B + A\overline{B}. \quad (18)$$

After completing the transistor-level schematic, a reusable XOR gate symbol was generated.

The physical layout was then developed using the required active, polysilicon, well, metal, contact, via, power, and ground layers.

The layout was checked using Design Rule Check and Layout Versus Schematic verification.

After successful physical verification, RC parasitic extraction was performed using the RCX flow.

The extracted circuit was then used for post-layout transient simulation so that the effects of layout-dependent parasitic resistance and capacitance could be included in the performance analysis.

The average post-layout power consumption obtained from the Cadence Virtuoso Calculator was approximately

$$P_{\text{avg}} = 13.2 \mu\text{W}. \quad (19)$$

The selected propagation-delay calculation produced the displayed value

$$t_p = -9.757 \text{ ns}. \quad (20)$$

The negative value suggests that the selected transition directions in the Calculator should be checked before using this value as the final physical propagation delay of the XOR gate.

15 Conclusion

A two-input CMOS XOR gate was successfully designed, implemented, and analyzed using Cadence Virtuoso.

The required Boolean function was

$$Y = A \oplus B. \quad (21)$$

The XOR gate produces a logic HIGH output when the two input signals are different and a logic LOW output when the two input signals are equal.

The project covered the major stages of the custom CMOS VLSI design flow, including schematic design, symbol generation, physical layout design, DRC verification, LVS verification, RC parasitic extraction, and post-layout simulation.

The physical layout was checked against the required fabrication design rules and its electrical connectivity was verified against the original schematic.

RCX was also completed successfully, allowing parasitic resistance and capacitance to be included in the final simulation.

The obtained post-layout average power consumption was

$$P_{\text{avg}} = 13.2 \mu\text{W} \quad (22)$$

and the selected Cadence Calculator expression displayed

$$t_p = -9.757 \text{ ns.} \quad (23)$$

The negative delay result should be checked by verifying the selected input and output threshold crossings in the Cadence Virtuoso Calculator.

Therefore, the project demonstrates the complete schematic-to-layout design, verification, extraction, and performance analysis of a two-input CMOS XOR gate using Cadence Virtuoso.

16 Software Used

The following software tools were used to complete the project:

- Cadence Virtuoso Schematic Editor
- Cadence Virtuoso Layout Editor
- Cadence Spectre Simulator
- Design Rule Check (DRC) Tool
- Layout Versus Schematic (LVS) Tool
- RCX / Parasitic Extraction Tool
- Cadence Virtuoso Visualization and Analysis Calculator